# 1. Velocity and Acceleration

## 1.1. Scalar vs. Vector Quantities

- Scalar quantities only have magnitude (do not have direction):
  - e.g., distance, speed, mass, and time (they all only have magnitude)
- Vector quantities have both magnitude and direction:
  - e.g., displacement, velocity, acceleration
  - can be positive or negative to show the direction

## 1.2. Kinematics Equations

(can only be used when acceleration is a constant)

$$v = u + at$$

$$s = ut + \frac{1}{2}at^2 \qquad and \qquad s = vt - \frac{1}{2}at^2$$

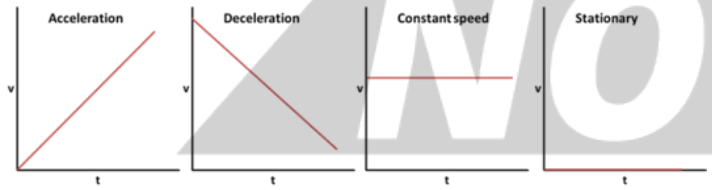
$$s = \frac{1}{2}(u+v)t$$

$$v^2 = u^2 + 2as$$

## 1.3. Displacement-Time Graph

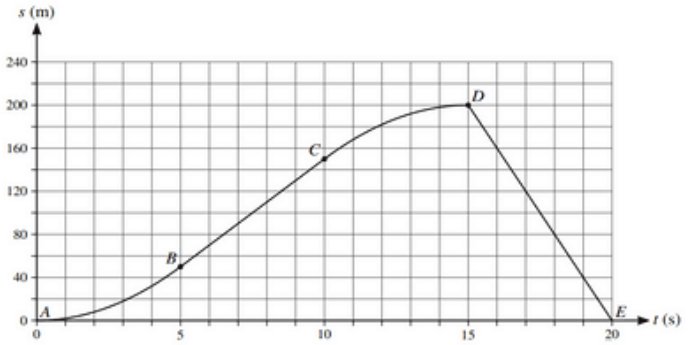
- Gradient = velocity

## 1.4. Velocity-Time Graph




- Gradient = acceleration
- Area under graph = change in displacement

{S22-P43} Question 3:




The displacement of a particle moving in a straight line is $s$ metres at time $t$ seconds after leaving a fixed point $O$. The particle starts from rest and passes through points $P$, $Q$ and $R$, at times $t = 5$, $t = 10$ and $t = 15$ respectively, and returns to $O$ at time $t = 20$. The distances $OP$, $OQ$ and $OR$ are 50 m, 150 m and 200 m respectively.

The diagram shows a displacement-time graph which models the motion of the particle from $t = 0$ to $t = 20$. The graph consists of two curved segments $AB$ and $CD$ and two straight line segments $BC$ and $DE$.

**(a)** Find the speed of the particle between $t = 5$ and $t = 10$. [1]

solution (a):

speed is the gradient of a distance-time graph. The line between $t = 5$ and $t = 10$ is linear and so it has a constant speed. The gradient is $= \frac{\Delta y}{\Delta x} = \frac{150-50}{10-5} = \frac{100}{5} = 20$ . So the speed of the particle between $t = 5$ and $t = 10$ is $20\ ms^{-1}$

**(b)** Find the acceleration of the particle between $t = 0$ and $t = 5$, given that it is constant. [2]

solution (b):

Here lets look at what we are given first. We have time (5 seconds), we have initial velocity (as the particle starts from rest, initial velocity = 0), and final velocity (which is the answer to the previous question, as we are trying to find the gradient of the graph at point B)

$t = 5\ \ s$, $u = 0\ \ ms^{-1}$ , $v = 20\ \ ms^{-1}$

we can use the kinematic equation $v = u + at$ (as acceleration is constant). $a = \frac{v-u}{t} = \frac{20-0}{5} = 4 \quad ms^{-2}$

So the acceleration of the particle between $t = 0$ and $t = 5$ is $= 4\ ms^{-2}$

**(c)** Find the average speed of the particle during its motion. [2]

solution (c)

the particle passes through the point R at $t = 15$ , which is a distance of 200m from the starting point. after reaching point R, the particle travels a further 200m from $t = 15$ to $t = 20$ . So the total distance traveled is 400m, which took 20 seconds to cover. avg speed = total distance/ total time = $\frac{400}{20} = 20$

So the average speed of the particle during its motion is $20\ ms^{-1}$

## 1.5. Average Velocity

- For an object moving with constant acceleration over a period of time, these quantities are equal:
  - The average velocity
  - The mean of initial & final velocities
  - Velocity when half the time has passed



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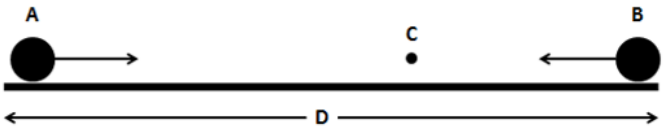
## 1.6. Relative Velocities




- Let $s_A$ be the distance travelled by $A$ and $s_B$ for $B$

$$s_A = ut + \frac{1}{2}at^2 \qquad s_B = ut + \frac{1}{2}at^2$$

- If a collision occurs at point $C$

$$s_A + s_B = D$$

- This gives you the time of when the collision occurred
- Same analysis if motion is vertical



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